WEEK 5 REPORT

Healthcare Data Cleaning & Data Modeling

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Tool Used: Microsoft Power BI Desktop

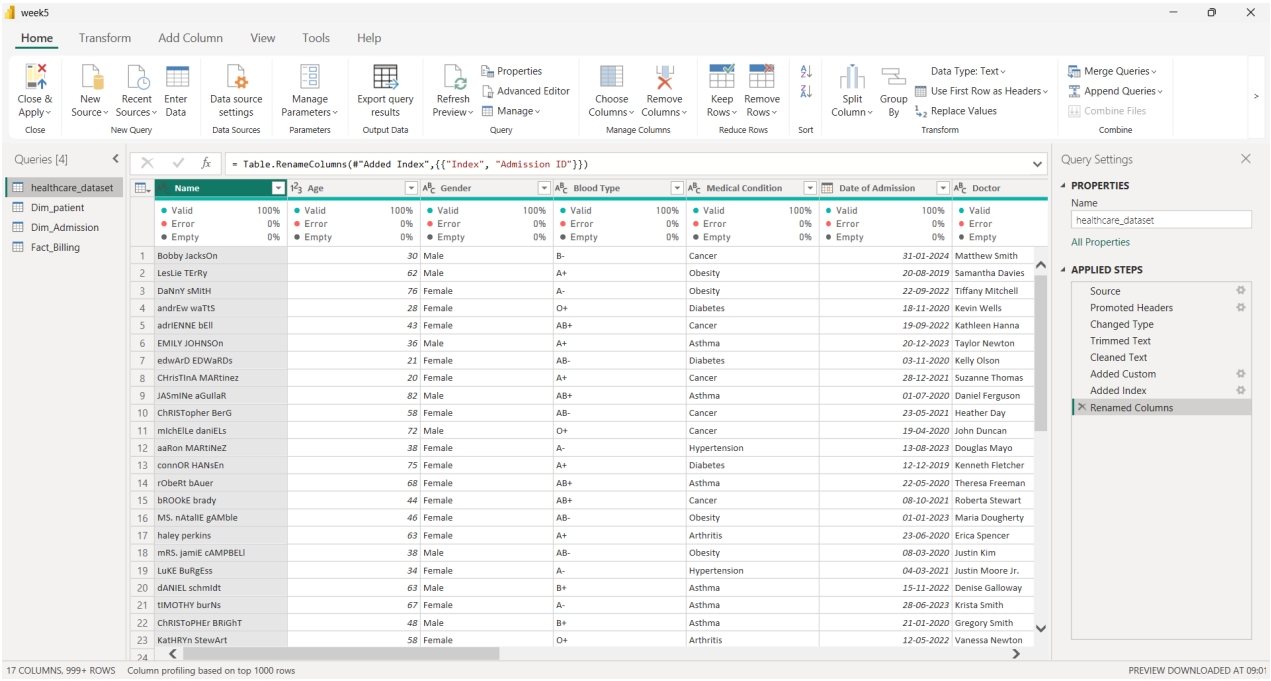
1. Introduction

This report documents the Week 5 Power BI task on Healthcare Data Cleaning & Data Modeling. The aim was to prepare patient and admission information, create age groups, organize the data into separate tables, and build an interactive dashboard for healthcare billing analysis.

2. Dataset Overview

The source file, healthcare_dataset.csv, includes Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Billing Amount, Room Number, Admission Type, Discharge Date, Medication, and Test Results. The source dataset was imported into Power Query for inspection.

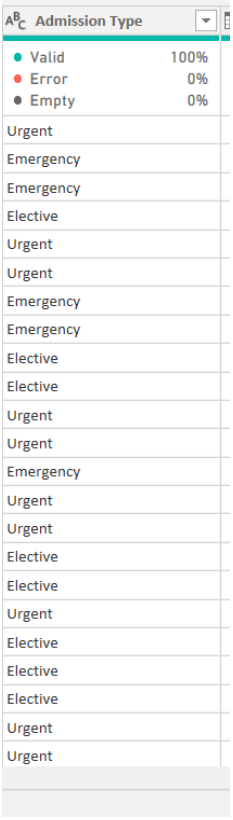
Source healthcare dataset and Power Query preparation steps.



3. Data Cleaning in Power Query

The source dataset was inspected in Power Query. In the captured profiling view, the visible columns showed 100% valid values, 0% errors, and 0% empty values. Trim and Clean were applied to categorical fields including Gender, Blood Type, Medical Condition, Admission Type, Medication, Test Results, and Insurance Provider. Admission Type was checked and showed Emergency, Elective, and Urgent.

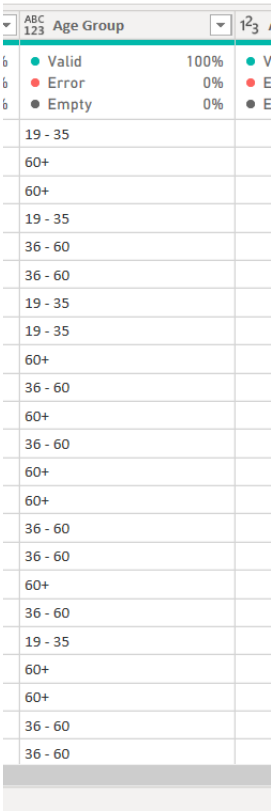
Admission Type categories and column quality.



4. Creating Age Groups

A custom Age Group column was created from Age using four buckets: 0–18, 19–35, 36–60, and 60+. The captured Power Query view showed 100% valid values for this column.

Age Group values in the prepared dataset.



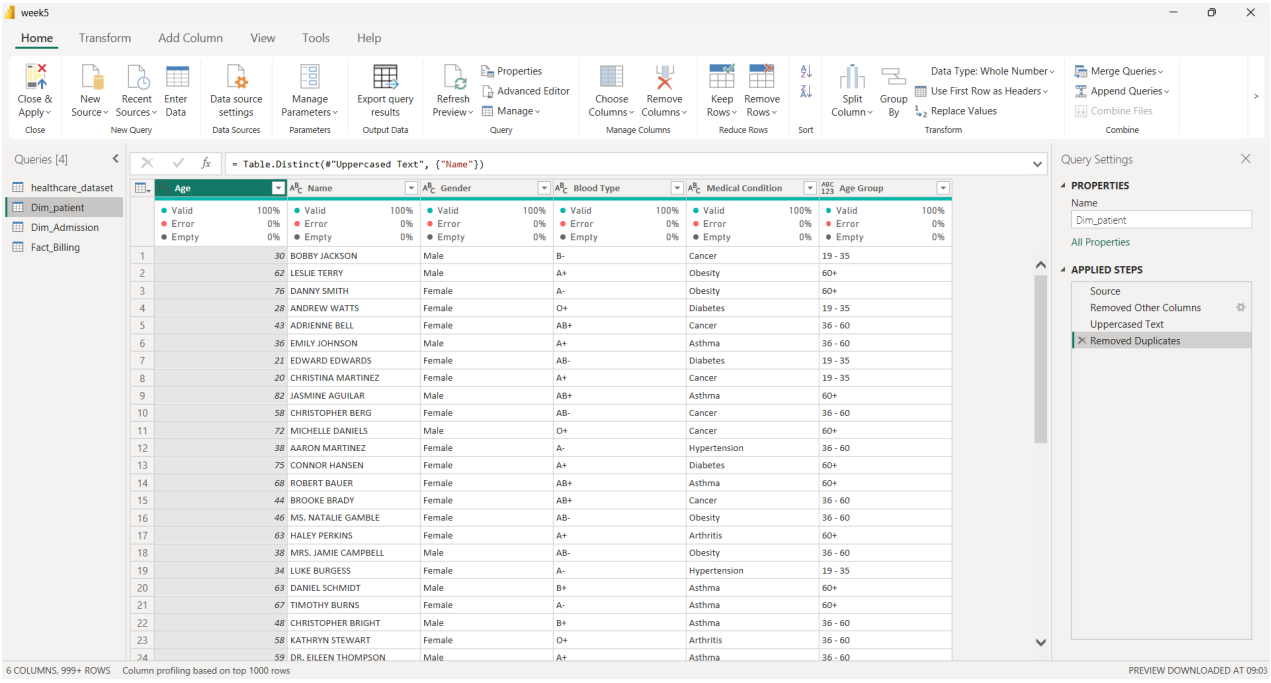
5. Admission ID

An index column was added to the source query and renamed Admission ID. This identifier was carried into Dim_Admission and Fact_Billing.

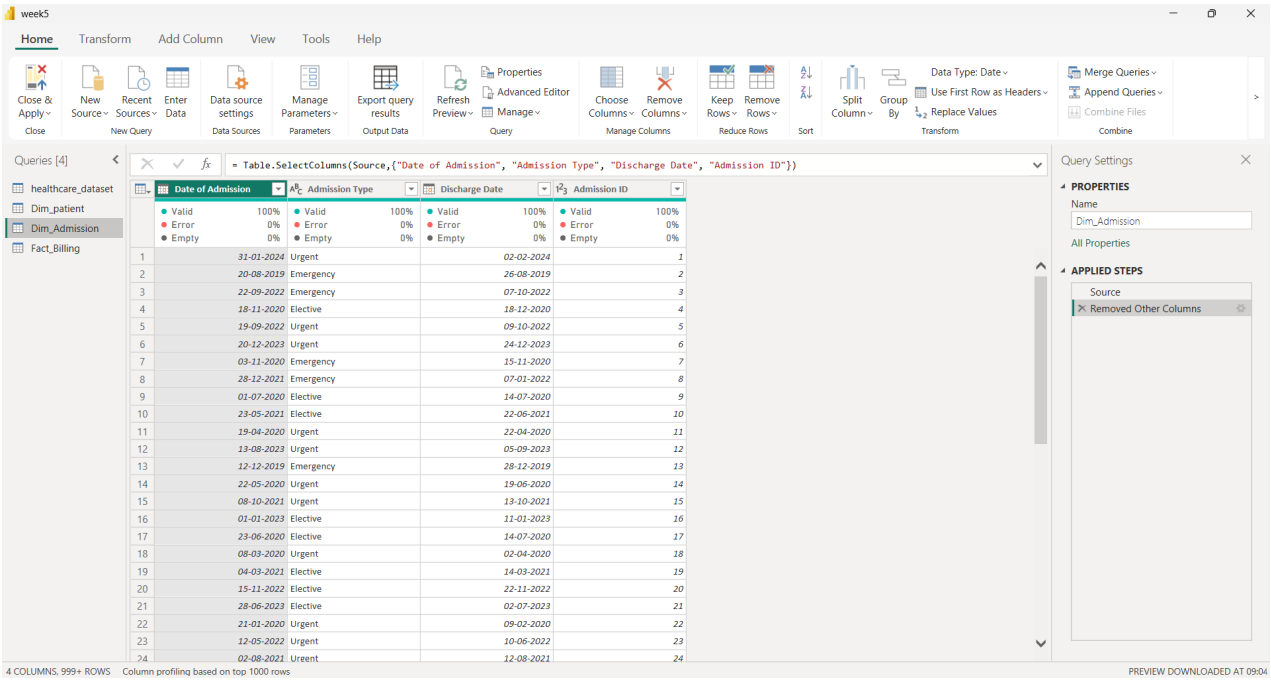
6. Data Model Tables

Dim_patient contains Name, Age, Gender, Blood Type, Medical Condition, and Age Group. Dim_Admission contains Admission ID, Date of Admission, Admission Type, and Discharge Date. Fact_Billing contains Admission ID, Name, Date of Admission, Insurance Provider, Billing Amount, and Room Number.

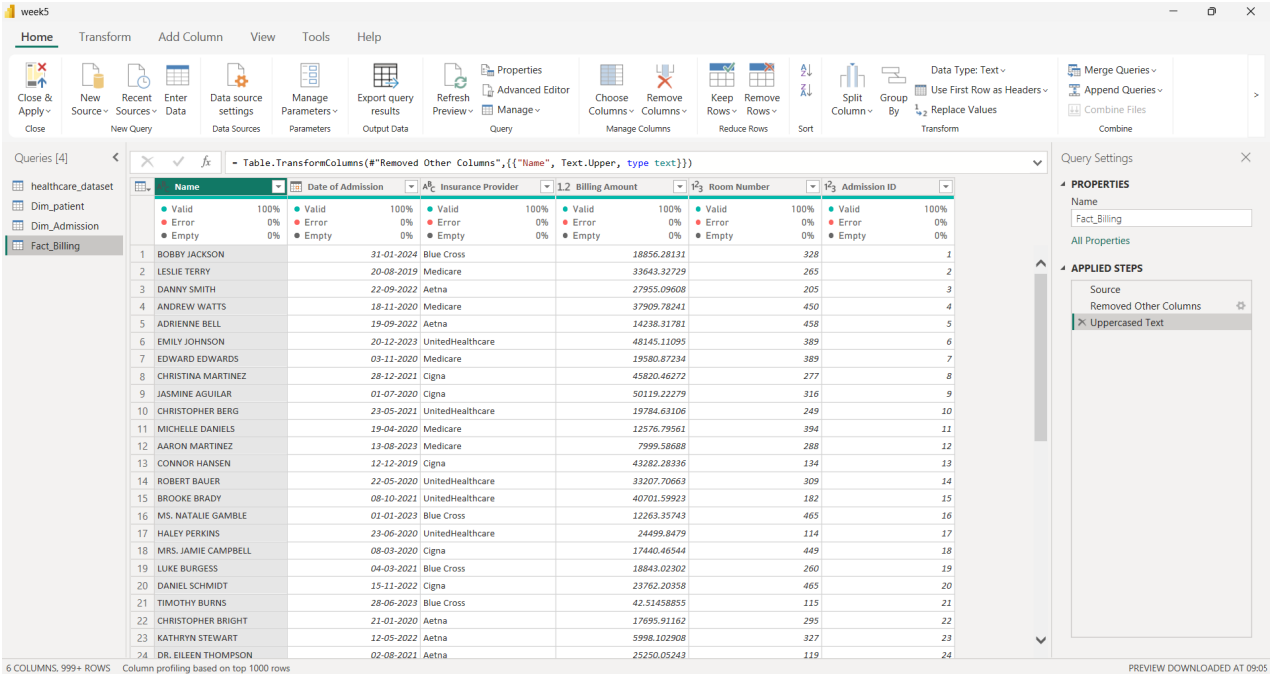
Dim_patient table with patient attributes and Age Group.



Dim_Admission table with Admission ID and admission dates.



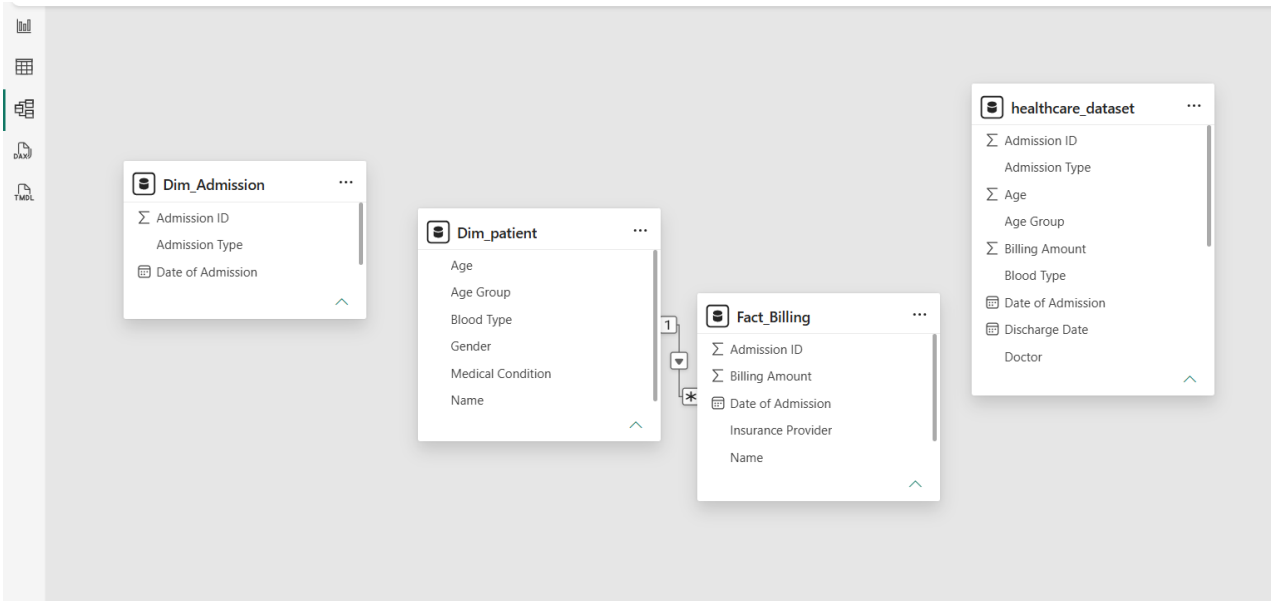
Fact_Billing table with billing details and Admission ID.



7. Relationships and Model View

The captured model view shows a one-to-many relationship from Dim_patient to Fact_Billing, with Dim_patient on the one side and Fact_Billing on the many side. The screenshots also show Dim_Admission and healthcare_dataset in the model. Dim_Admission is not visibly connected in the captured model view, so this report does not assume an additional relationship.

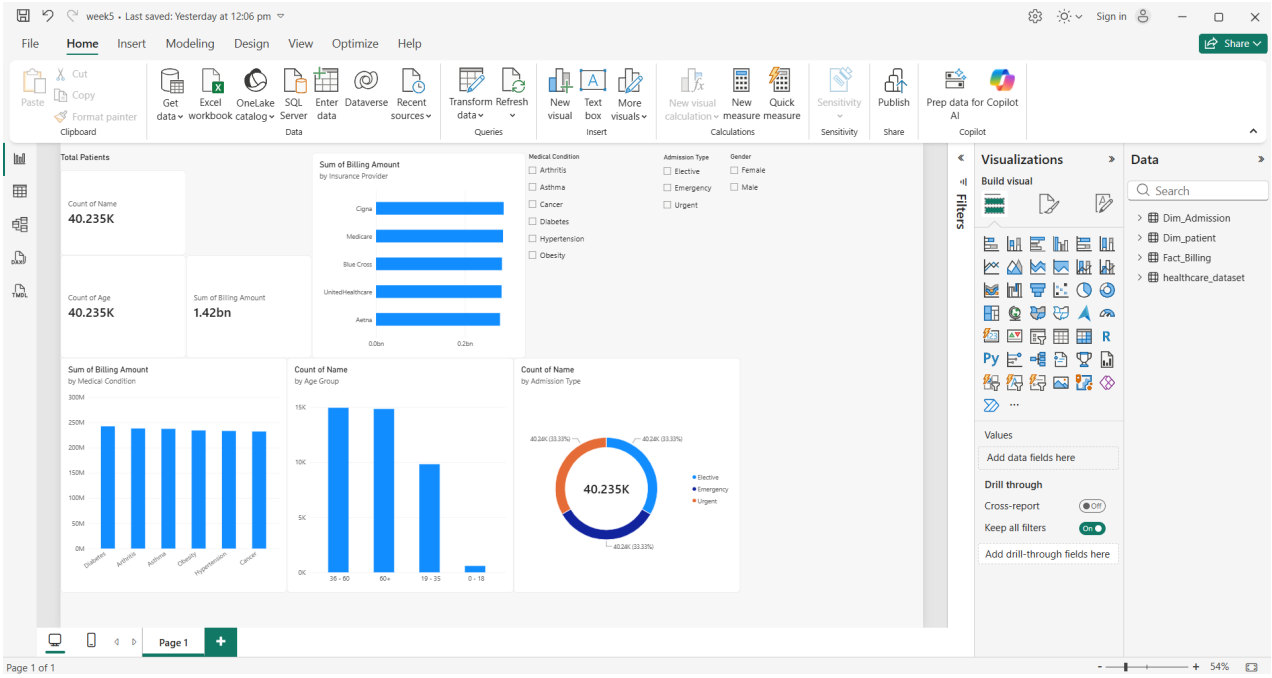
Model view showing the patient-to-billing relationship.



8. Dashboard and Visualizations

The dashboard includes cards for Count of Name (40.235K), Count of Age (40.235K), and Sum of Billing Amount (1.42bn). It also includes billing by insurance provider, billing by medical condition, counts by age group, and counts by admission type. Slicers are provided for Gender, Medical Condition, and Admission Type.

Final healthcare dashboard with cards, charts, and slicers.



9. Observations

The dashboard allows comparisons of billing totals across insurance providers and medical conditions, shows the distribution of records across age groups, and supports filtering by gender, medical condition, and admission type.

10. Conclusion

The healthcare dataset was prepared in Power Query, age groups and an Admission ID were created, and separate patient, admission, and billing tables were formed. A Power BI dashboard was built using cards, charts, and slicers to summarize the prepared healthcare data.